

INTRO TO SYNTHESIS

A crash course in subtractive synthesis, on the Sequential Fourm





TODAY

Club Control, Bucharest

SINTEZAUR

Bucharest's community for people who love synthesis. Your guide today: Iulian.

ZEEDO

Our retail partner, and the reason there are a dozen synths in this room.

SEQUENTIAL FOURM

Today's instrument: a 4-voice analog synth that feels pressure in every key.

SINTEZAUR × ZEEDO · POWERED BY SEQUENTIAL

WHERE ON FOURM

TRANSPOSE DOWN + PROGRAM

Hold TRANSPOSE DOWN (left of the keys) and press PROGRAM: the synth resets to the Basic Program.



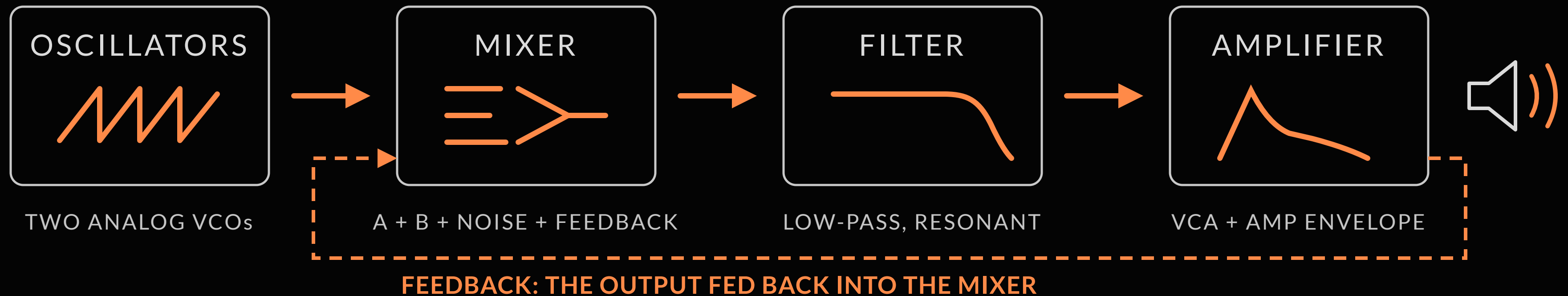
SIGNAL FLOW

How sound travels through a subtractive synthesizer



THE VOICE PATH

Oscillators, mixer, filter, amplifier: one road for every sound



WHERE ON FOURM

The signal path, left to right across the panel



THE MIXER

Every voice starts here: four sources feed the filter

OSC A



SAW + PULSE
THE BRIGHT BACKBONE

OSC B



TRI + SAW + PULSE
WEIGHT, DETUNE, LO MODES

NOISE



WHITE / PINK / VIOLET
AIR, PERCUSSION, TEXTURE

FEEDBACK



OUTPUT LOOP
SOFT-CLIP SATURATION

TURN UP AT LEAST ONE SOURCE, OR LET RESONANCE SELF-OSCILLATE

KEY TERMS

New words from this module

SIGNAL FLOW

The road a sound takes: oscillators → mixer → filter → amplifier → output.

VCO · OSCILLATOR

Voltage-controlled oscillator. Generates the raw waveform. Fourm has two.

MIXER

Sets the level of each source: Osc A, Osc B, noise, feedback.

HARMONICS

Whole-number multiples of the fundamental, stacked above it and usually fainter; their mix is timbre.

NOISE

Unpitched sound, in white, pink, or violet flavors.

FEEDBACK

The output fed back into the mixer, soft-clipped: warmth, then growl.

VCF · VCA

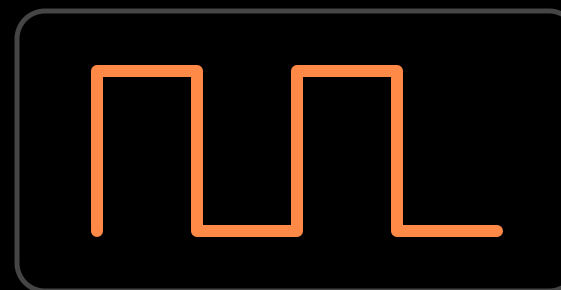
Voltage-controlled filter and amplifier: the tone and loudness stages.

OSCILLATORS

Where every sound is born



SAW



PULSE



TRI

THE WAVESHAPES

Two oscillators, five buttons, endless timbres



SAW: ALL HARMONICS · PULSE: ODD ONLY WHEN SQUARE



ADDS WEIGHT, DETUNE, LO MODES

LIGHT SEVERAL SHAPE BUTTONS: THE WAVES ADD UP

OCTAVES & TUNING

Detune B against A and the sound starts to breathe.

-2

-1

0

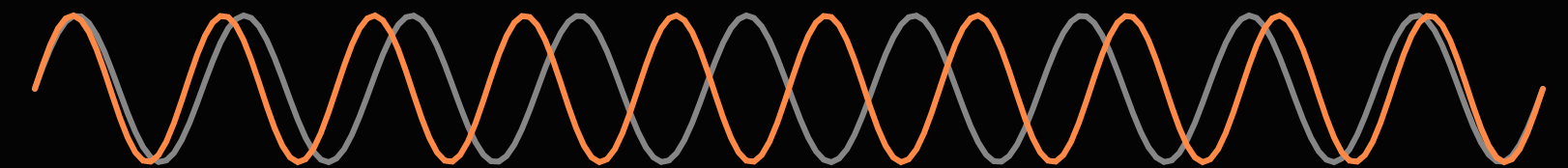
+1

+2

FIVE OCTAVE POSITIONS ON EACH OSCILLATOR

OSC B ONLY: LO1 · LO2, BELOW THE OCTAVES —
SUB-AUDIO, IT BECOMES A MODULATOR (LEVEL 2)

FREQUENCY · FINE TUNE · ± 7 SEMITONES



A AND B, A FEW CENTS APART



TOGETHER THEY BEAT: THE SOUND BREATHES

TOO FAR AND YOU HEAR TWO NOTES: BRING IT BACK UNTIL IT JUST SHIMMERS

WHERE ON FOURM

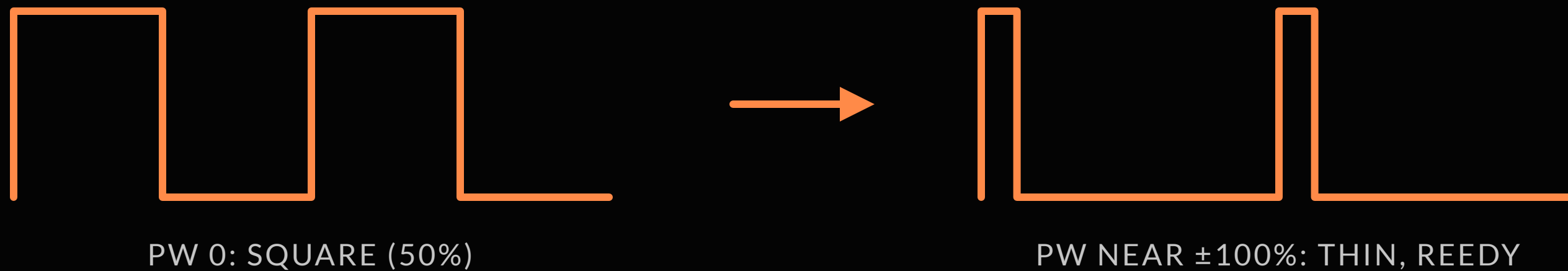
OSC B › OCTAVE · FREQUENCY

OCTAVE is the encoder. Turn FREQUENCY slowly off centre: slow beats, chorus, two notes, then back.



PULSE WIDTH & PWM

No PW knob: hold PULSE, turn SELECT. 0 = square. Toward ± 100 : thin, then silence.



AT FULL $\pm 100\%$ THE PULSE NARROWS TO SILENCE

ROUTE LFO OR FILT ENV TO PW: THAT MOVEMENT IS PWM

WHERE ON FOURM

OSC A › PULSE|PW + SELECT

Hold PULSE|PW and turn SELECT: zero is the square, towards the edge the pulse thins to a thread.



THE SUB TRICK

Osc B, one octave down, triangle

OSC A: SAW · OCTAVE 0



OSC B: TRI · OCTAVE -1



TOGETHER



THE TRIANGLE ADDS A CLEAN OCTAVE BELOW: WEIGHT WITHOUT MUD

KEY TERMS

New words from this module

WAVESHAPE

The geometric shape of one cycle: it sets the raw timbre.

PWM

Pulse width modulation: moving PW with the LFO or an envelope.

LO1 / LO2

Osc B's low modes: LFO-range pitch, key-tracked (LO1) or free (LO2).

PULSE WIDTH (PW)

Duty cycle of the pulse wave, -100% to +100%. 0 = square.

SYNC

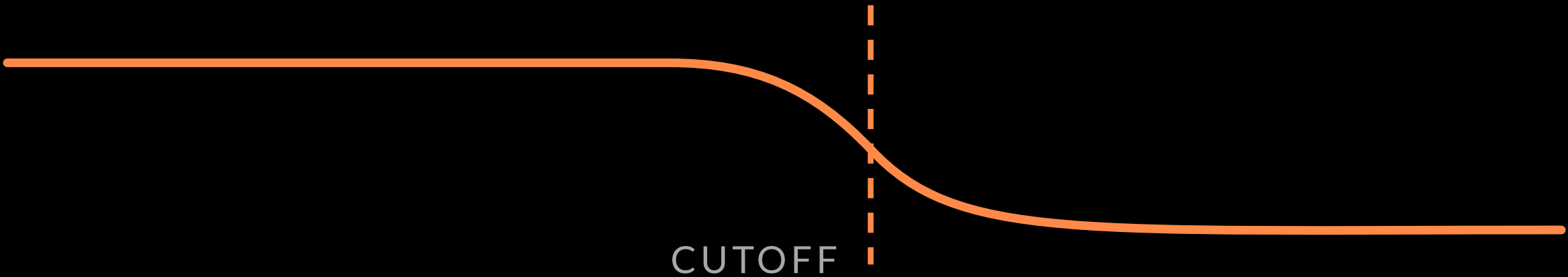
Locks Oscillator A's cycle to Oscillator B for aggressive timbres.

HARMONICS

Saw has them all; square and triangle only the odd ones, the triangle's fade much faster.

FILTER

Shaping your tone



FILTER

ENV AMT

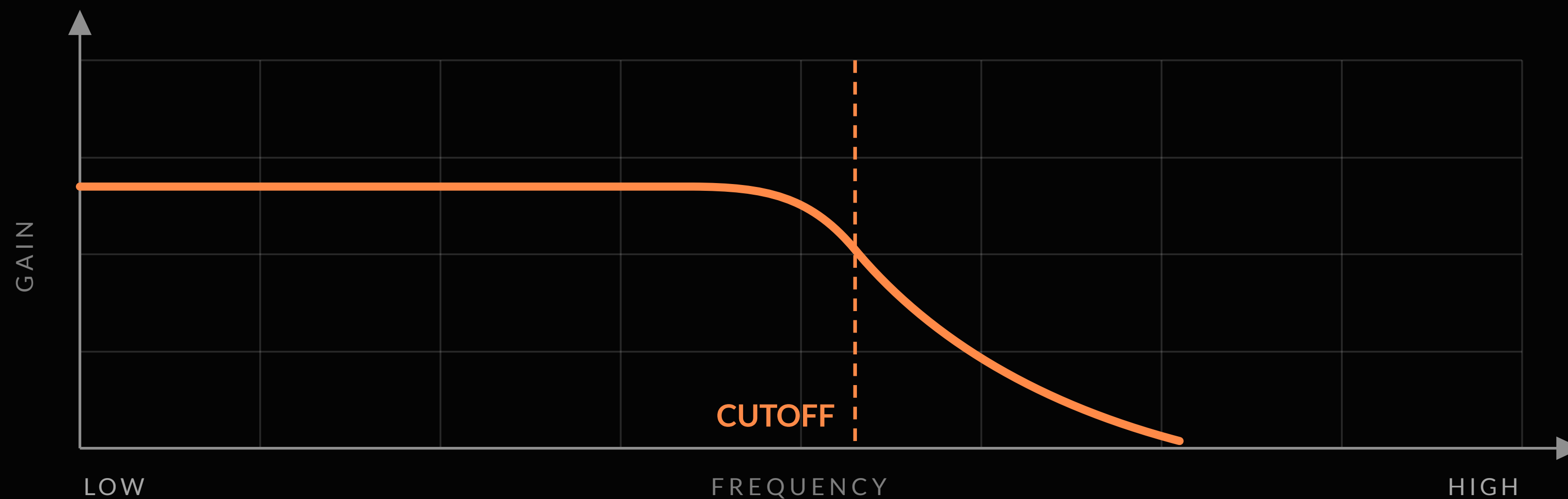
KEY TRACK AMOUNT

CUTOFF

RESONANCE

LOW-PASS

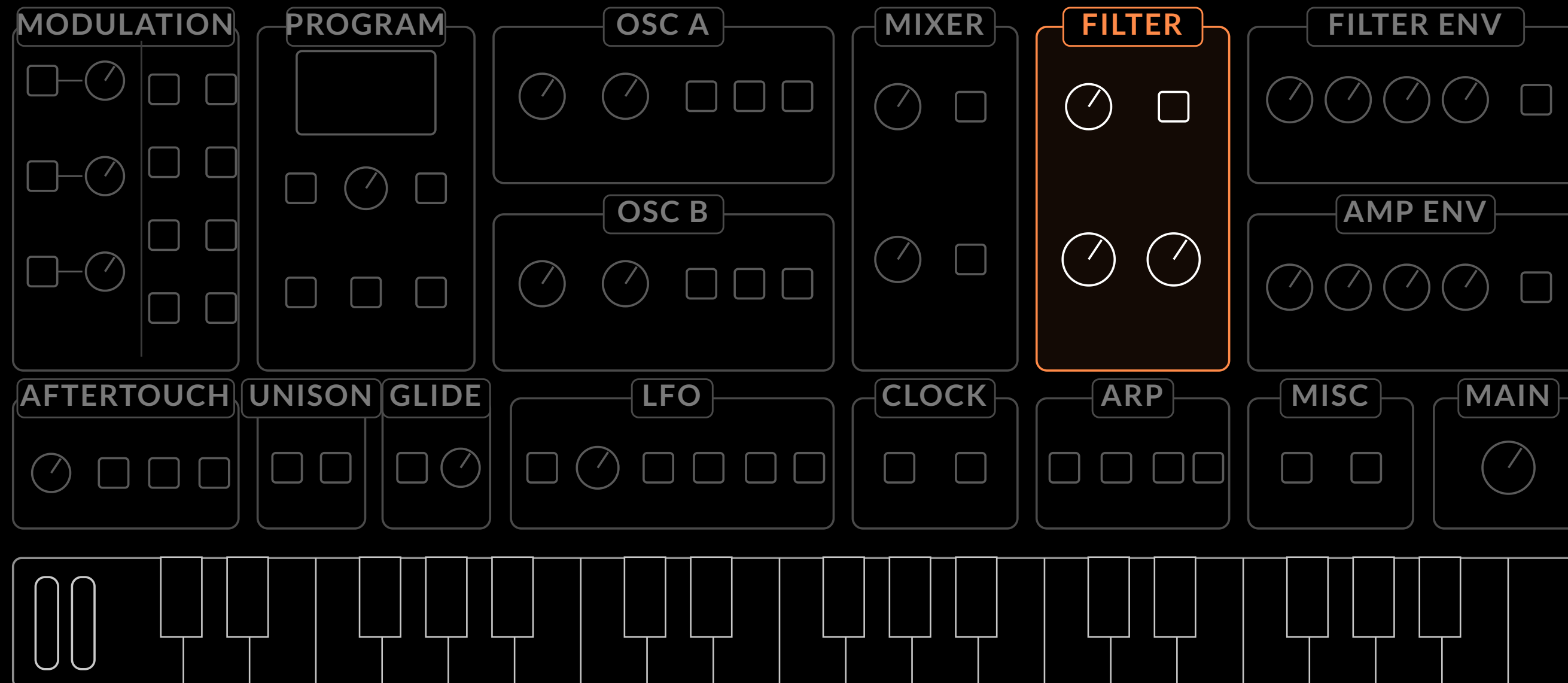
Below the cutoff passes, above it fades. Close it: darker. Open it: brighter.



WHERE ON FOURM

FILTER › CUTOFF

Close CUTOFF slowly, then open it again: the harmonics leave and come back one by one.



RESONANCE

A boost right at the cutoff. Turn it up and the filter sings; at maximum it becomes a sound source of its own.



WHERE ON FOURM

FILTER › RESONANCE

RESONANCE up while you sweep CUTOFF: a peak that whistles along; all the way up the filter self-oscillates: it sounds on its own.



KEY TERMS

New words from this module

CUTOFF

The hinge: below passes, above fades.

SELF-OSCILLATION

At maximum, the filter becomes a sine source.

ENV AMOUNT

How much the envelope moves the cutoff.

RESONANCE

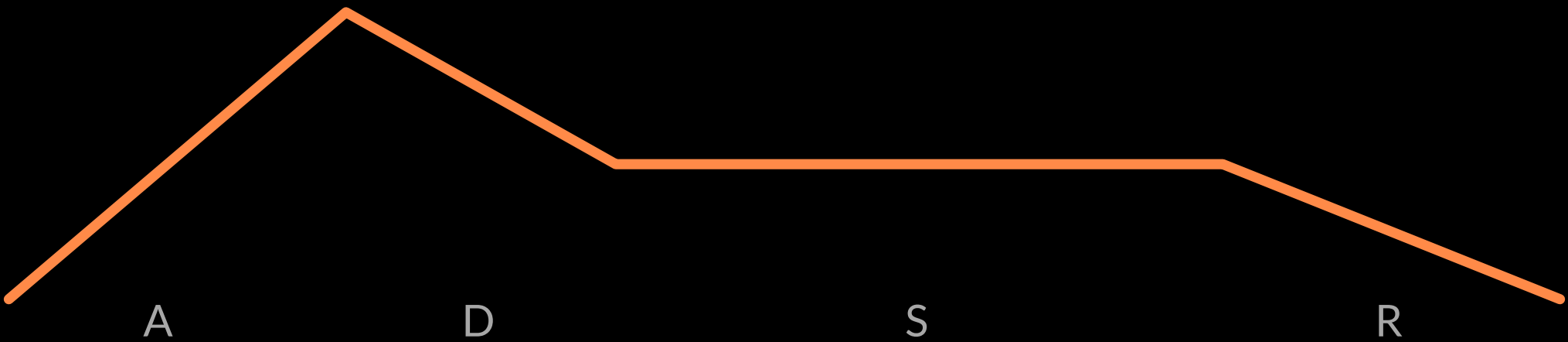
A boost right at the cutoff.

KEY TRACK

Higher notes, brighter filter.

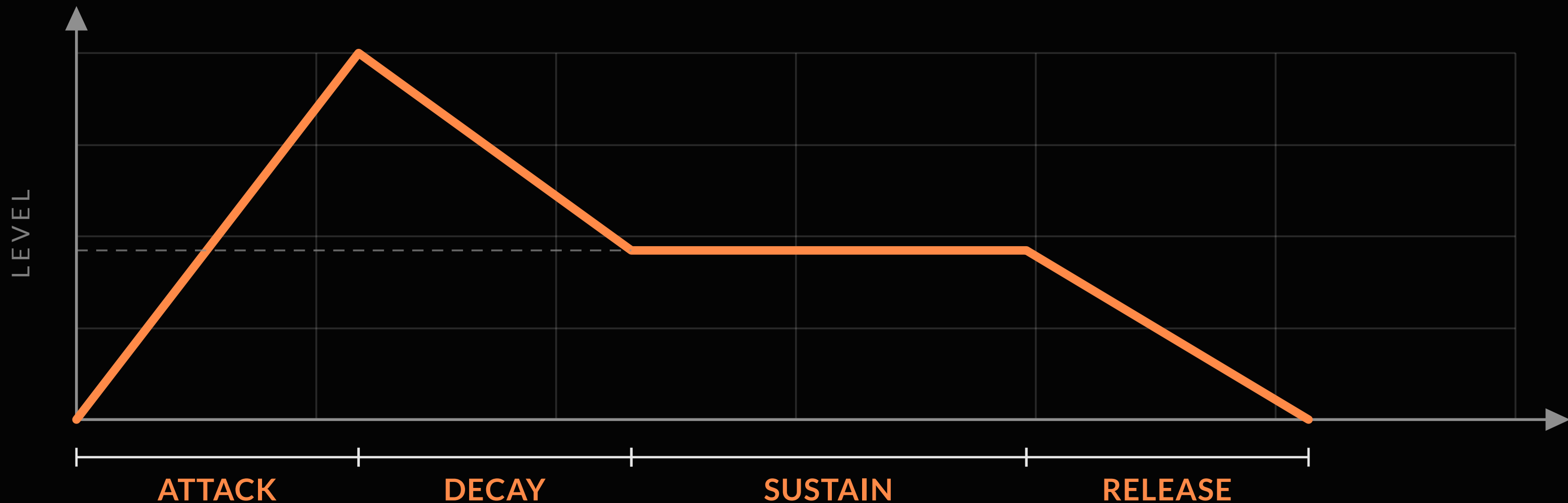
ENVELOPES

Controlling your sound over time



ADSR

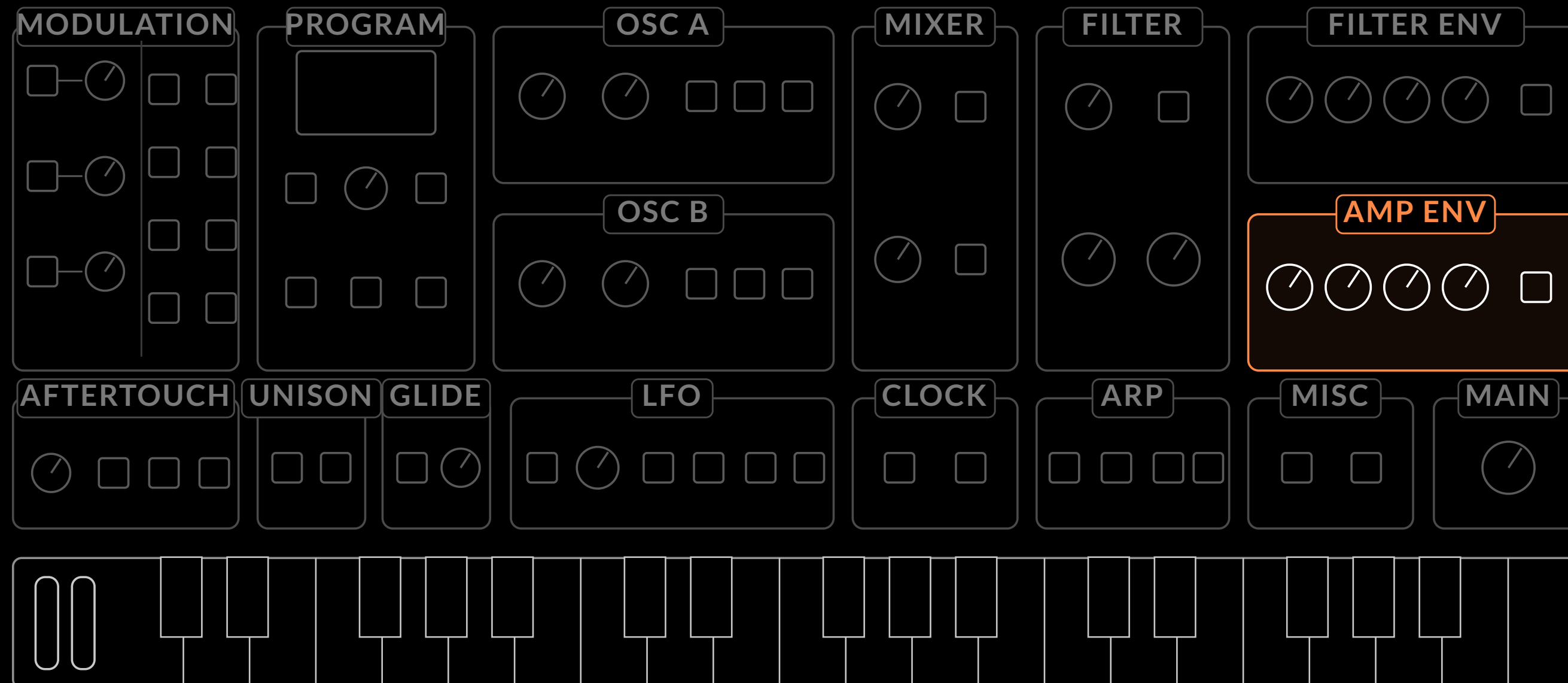
A shape that happens every time you press a key. Sustain is a level, not a time.



WHERE ON FOURM

AMP ENVELOPE › ATTACK · DECAY · SUSTAIN · RELEASE

Four knobs, one shape. Play a note, turn ATTACK up, play it again; then let go and hear RELEASE. The note learns to breathe.



TWO ENVELOPES

Same shape, two jobs: how loud, how bright.

AMP ENVELOPE



LOUDNESS
HOW LOUD

FILTER ENVELOPE

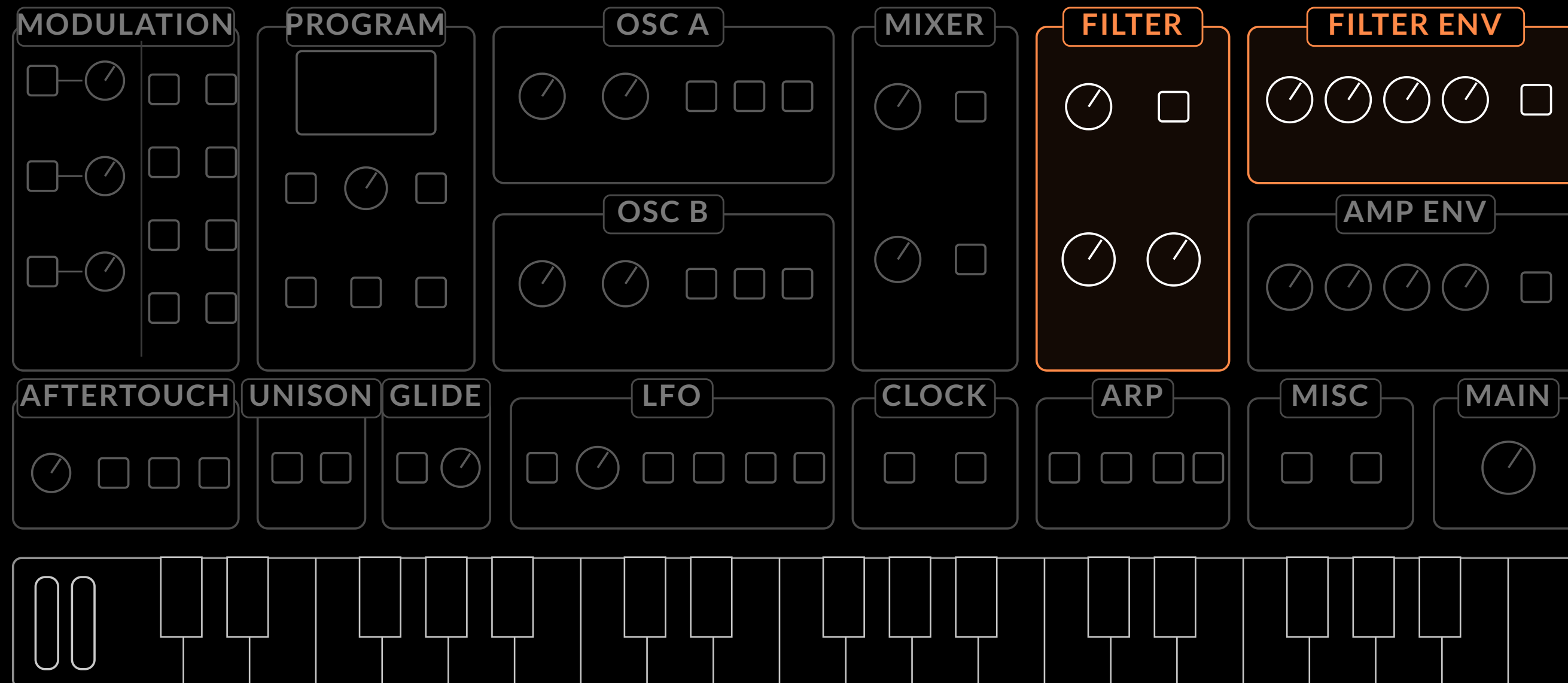


CUTOFF · SCALED BY ENV AMT
HOW BRIGHT

WHERE ON FOURM

FILTER ENV › DECAY + FILTER › ENV AMT

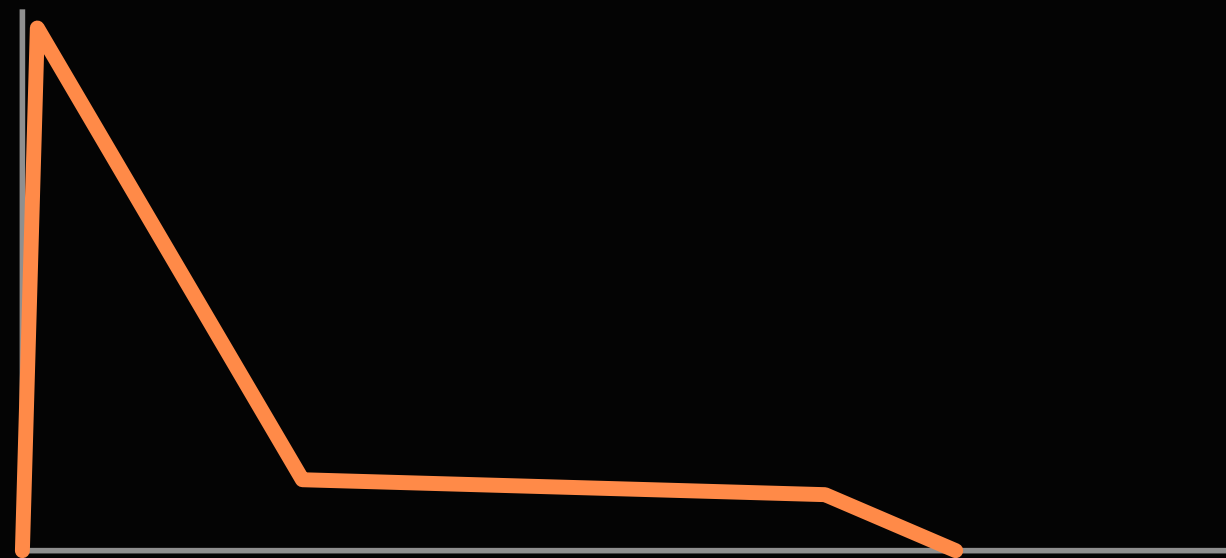
ENV AMT (in the filter) sends the envelope to the cutoff; DECAY sets how fast the brightness falls away.



BASS vs PAD

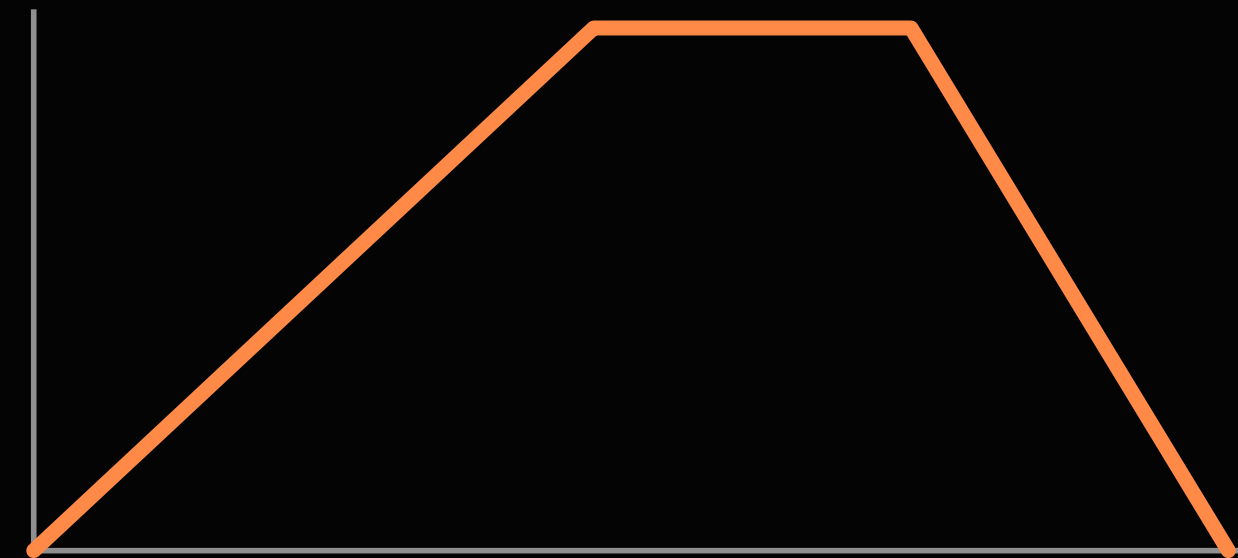
The same oscillators, two instruments.

BASS



INSTANT ATTACK · SHORT DECAY
LOW SUSTAIN

PAD



SLOW ATTACK · NO DECAY
FULL SUSTAIN · LONG RELEASE

KEY TERMS

New words from this module

ATTACK

Time from silence to the peak.

SUSTAIN

A level, not a time.

ENV AMOUNT

How much the envelope moves the cutoff.

DECAY

Time from the peak to the sustain level.

RELEASE

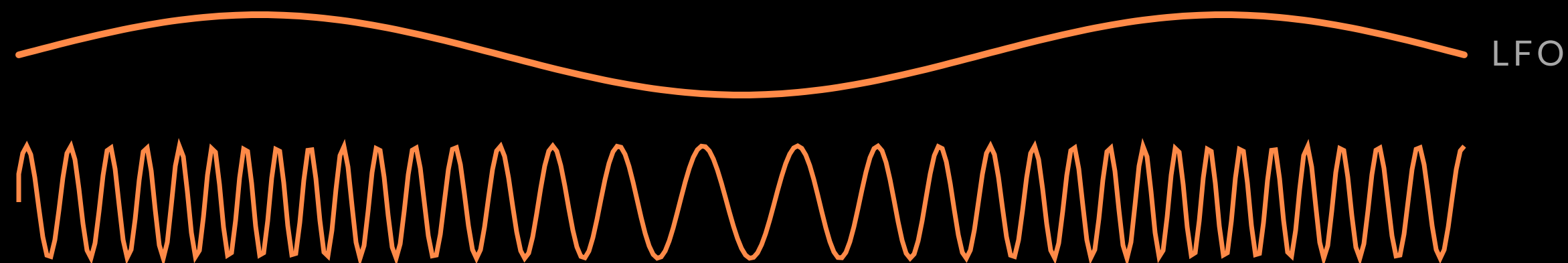
Time to fade after the key is let go.

VELOCITY

How hard you strike scales the envelope.

LFO & MODULATION

Adding movement



THE LFO

An oscillator too slow to hear. You hear what it moves.

LFO

→ PITCH **VIBRATO**



→ CUTOFF **WAH**



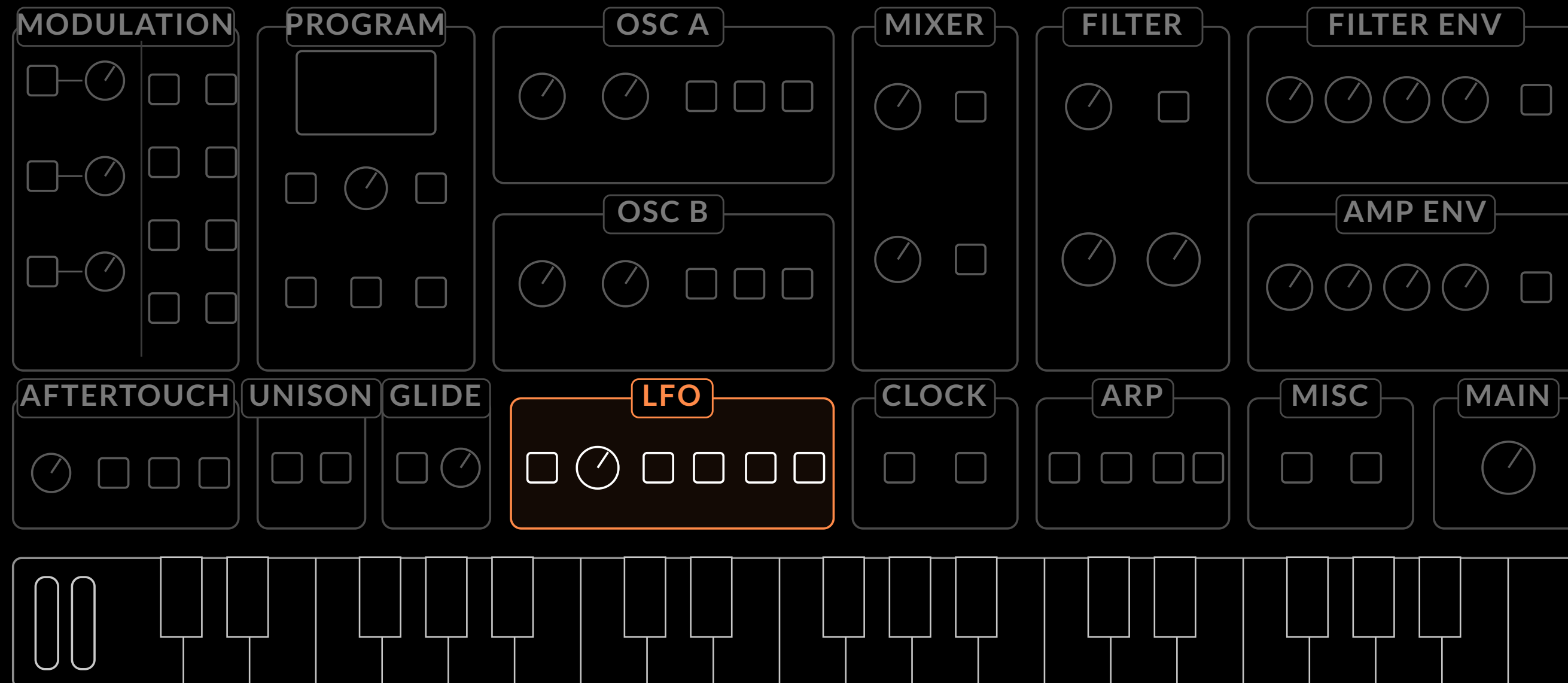
→ AMP **TREMOLO**



WHERE ON FOURM

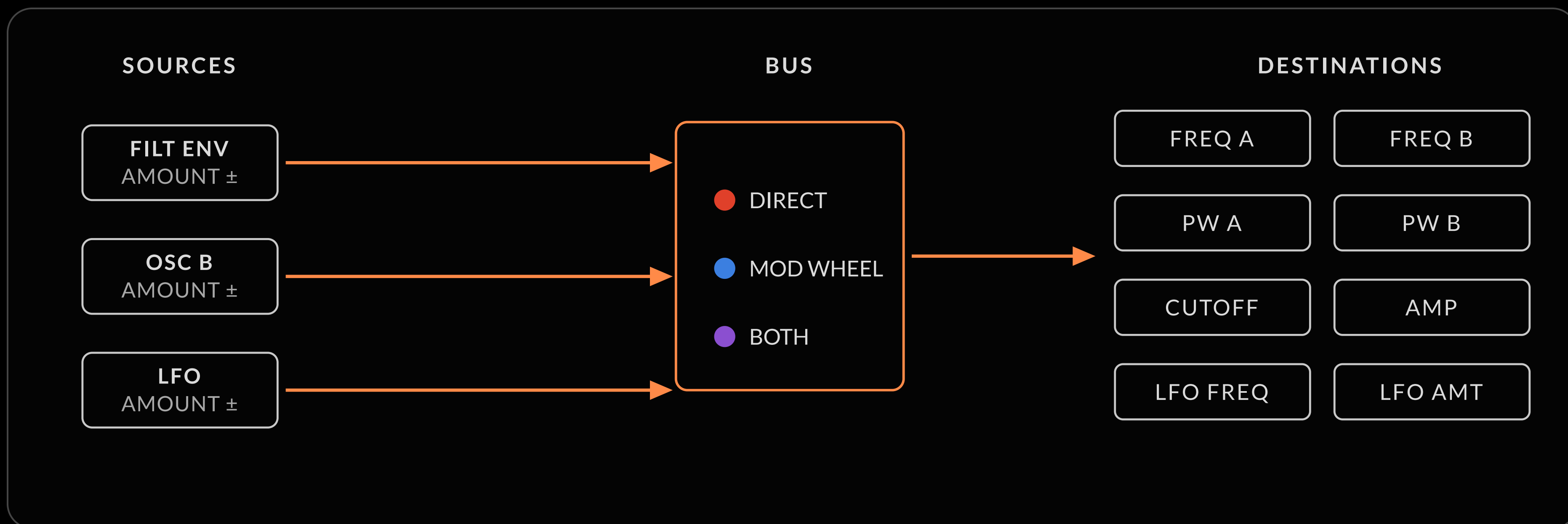
LOW FREQUENCY OSCILLATOR

Mod wheel up for vibrato: the Basic Program already wires it. FREQUENCY sets the speed, the shape buttons the motion; how far it goes is the LFO's AMOUNT knob in the MODULATION section, next slide.



THE MODULATION SECTION

Route → destination → amount. On the wheel bus, the mod wheel is the volume of the modulation.

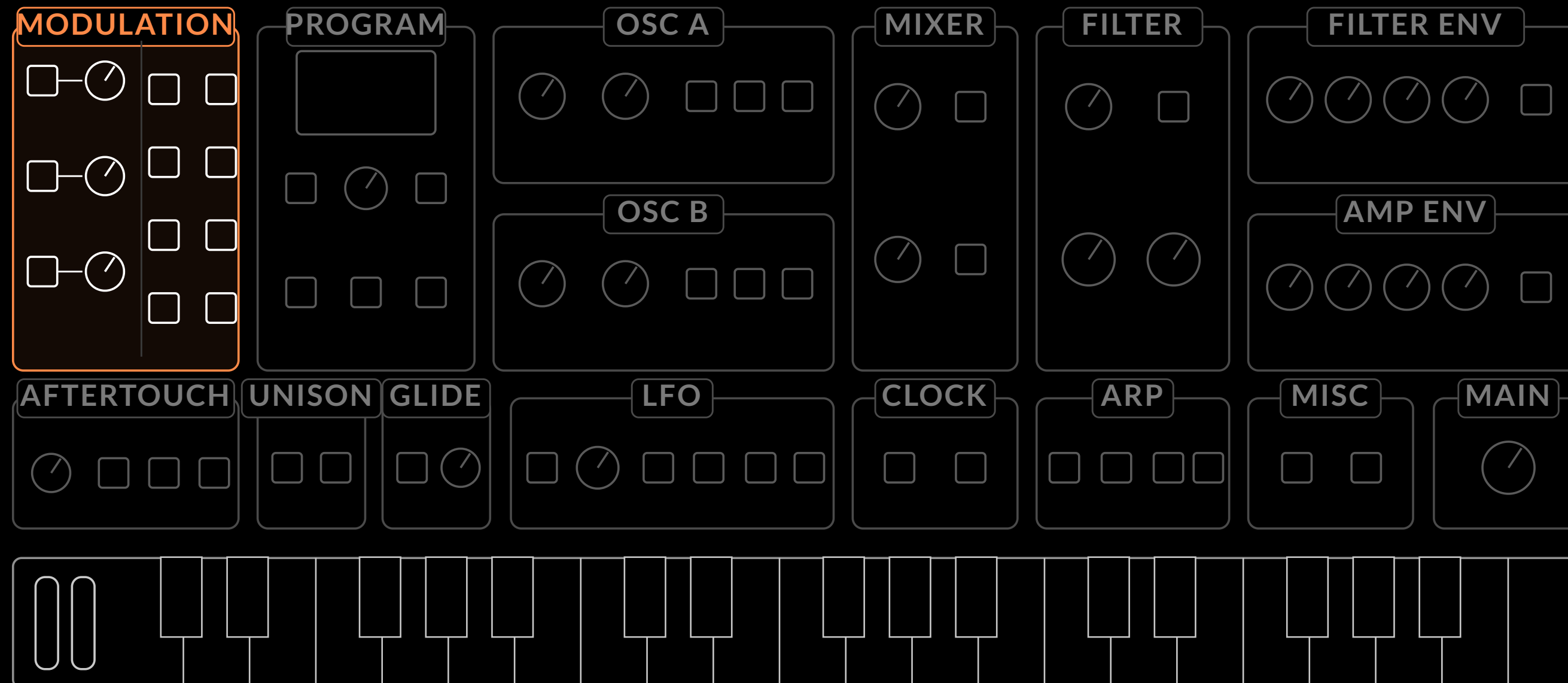


WHERE ON FOURM

MODULATION › SOURCE · DESTINATION · AMOUNT

Three sources. Press a ROUTE, then a destination, then AMOUNT: zero at 12 o'clock.

- red = the direct bus, always on
- blue = the mod-wheel bus
- purple = both



KEY TERMS

New words from this module

LFO

An oscillator too slow to hear.

VIBRATO

LFO to pitch.

PWM

Pulse width moved by a modulator, usually the LFO.

MOD WHEEL

The volume of the modulation.

RATE • FREQUENCY

Its speed (the FREQUENCY knob on Fourm); SYNC locks it to the clock.

TREMOLO

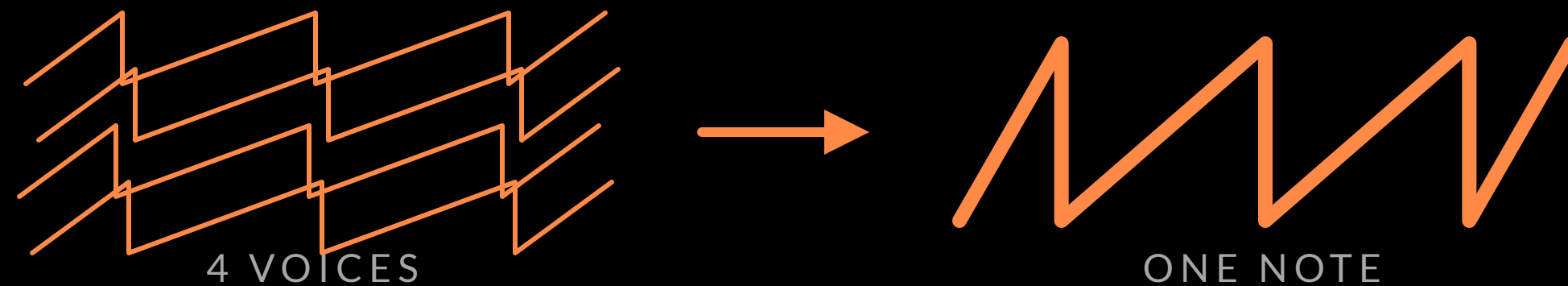
LFO to amplitude.

MODULATION BUS

Direct (red) or mod wheel (blue).

VOICES & UNISON

Four voices, one note, and the loop that thickens everything

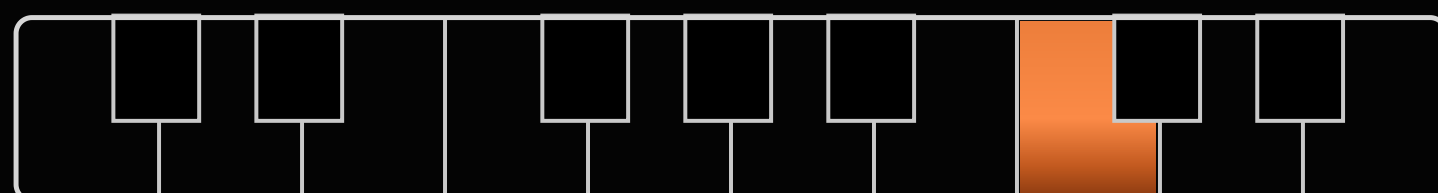


POLYPHONY

Fourm has four complete voices. Play a fifth note and the oldest one is stolen.

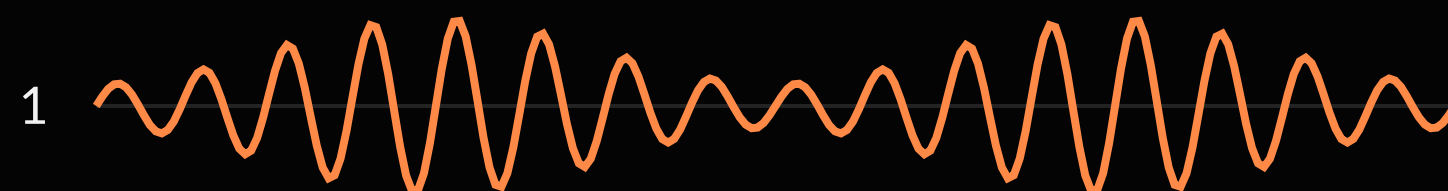
MONOPHONIC · ONE VOICE

WHAT YOU PLAY



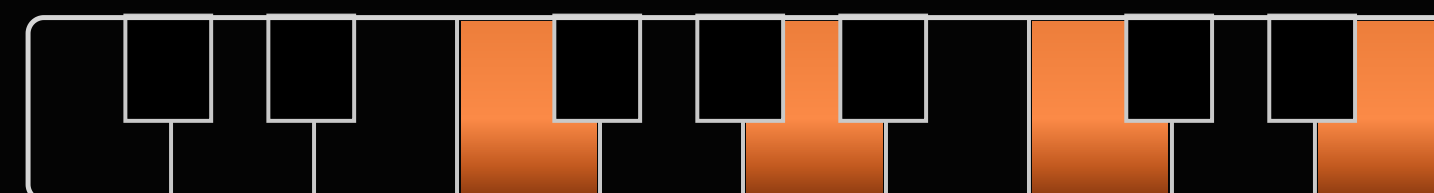
↓ LAST NOTE

WHAT YOU HEAR

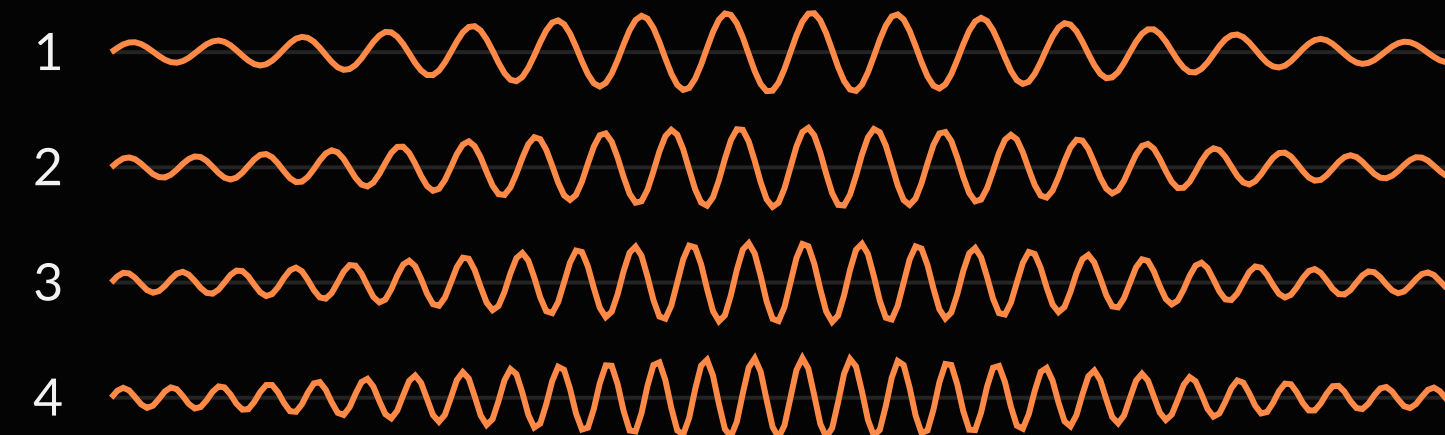


4-VOICE POLYPHONIC

WHAT YOU PLAY



WHAT YOU HEAR



UNISON

Four voices on one note



1-4 VOICES, SLIGHTLY DETUNED



ONE HUGE MONOPHONIC NOTE

HOLD UNISON + SELECT FOR VOICES • PARAM 10: UNISON DETUNE 0-127

WHERE ON FOURM

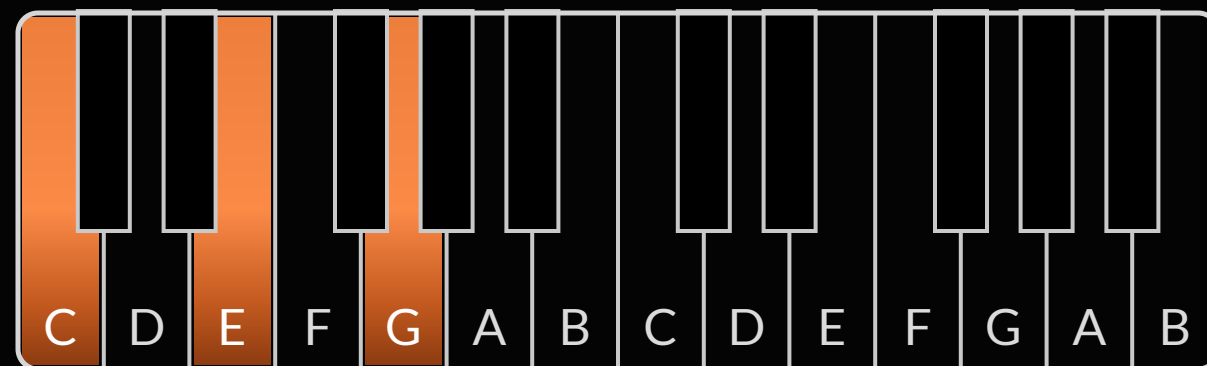
UNISON + SELECT · DISPLAY: PARAM 10

UNISON on; hold it and turn SELECT for the voice count. The detune between them lives at Param 10 on the display.

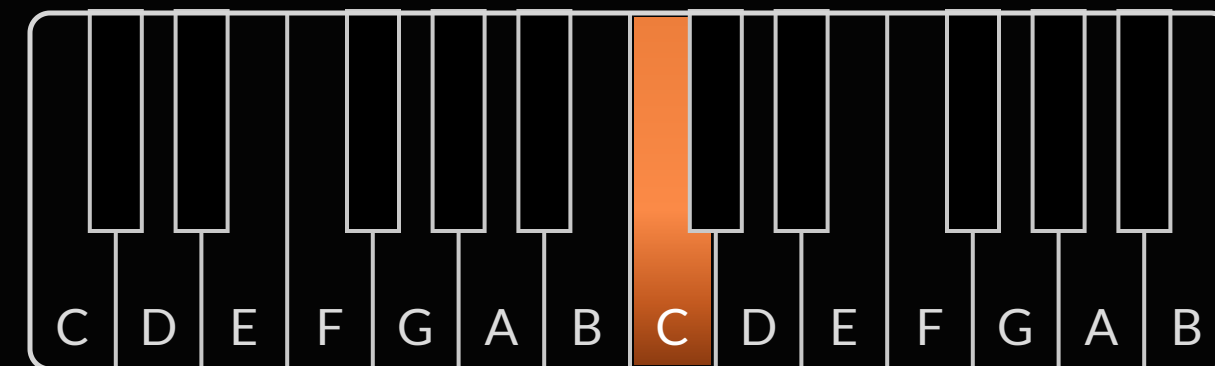


CHORD MEMORY

One key plays the whole chord



1 · HOLD A CHORD, PRESS UNISON



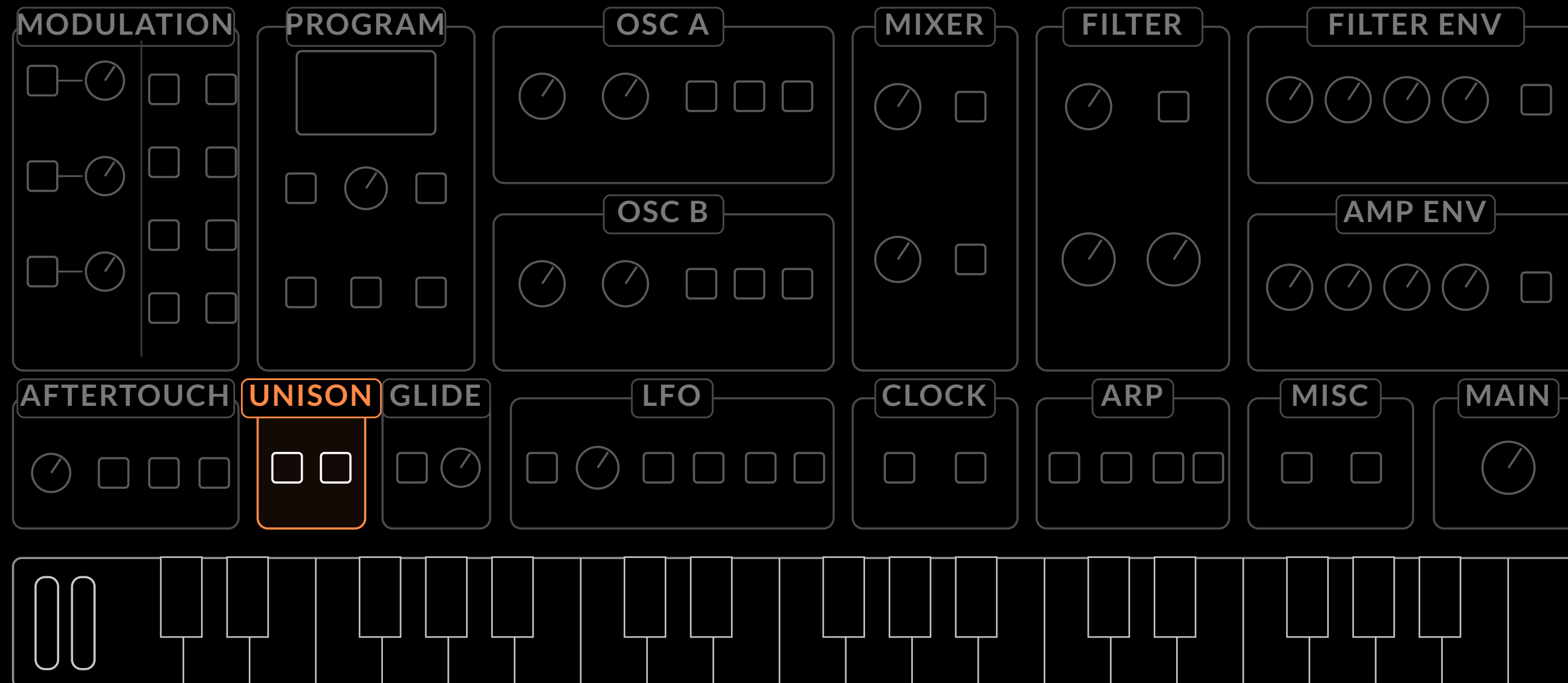
2 · ANY SINGLE KEY REPLAYS IT, TRANSPOSED

SAVED WITH THE PROGRAM: GREAT FOR CHORD STABS

WHERE ON FOURM

UNISON › CHORD MEMORY

With Chord selected in the UNISON menu, hold a chord and press UNISON: from now on any single key plays the whole chord, transposed. Full steps: handbook, module 06.



FEEDBACK

The output fed back into the mixer, soft-clipped: warmth, then growl.

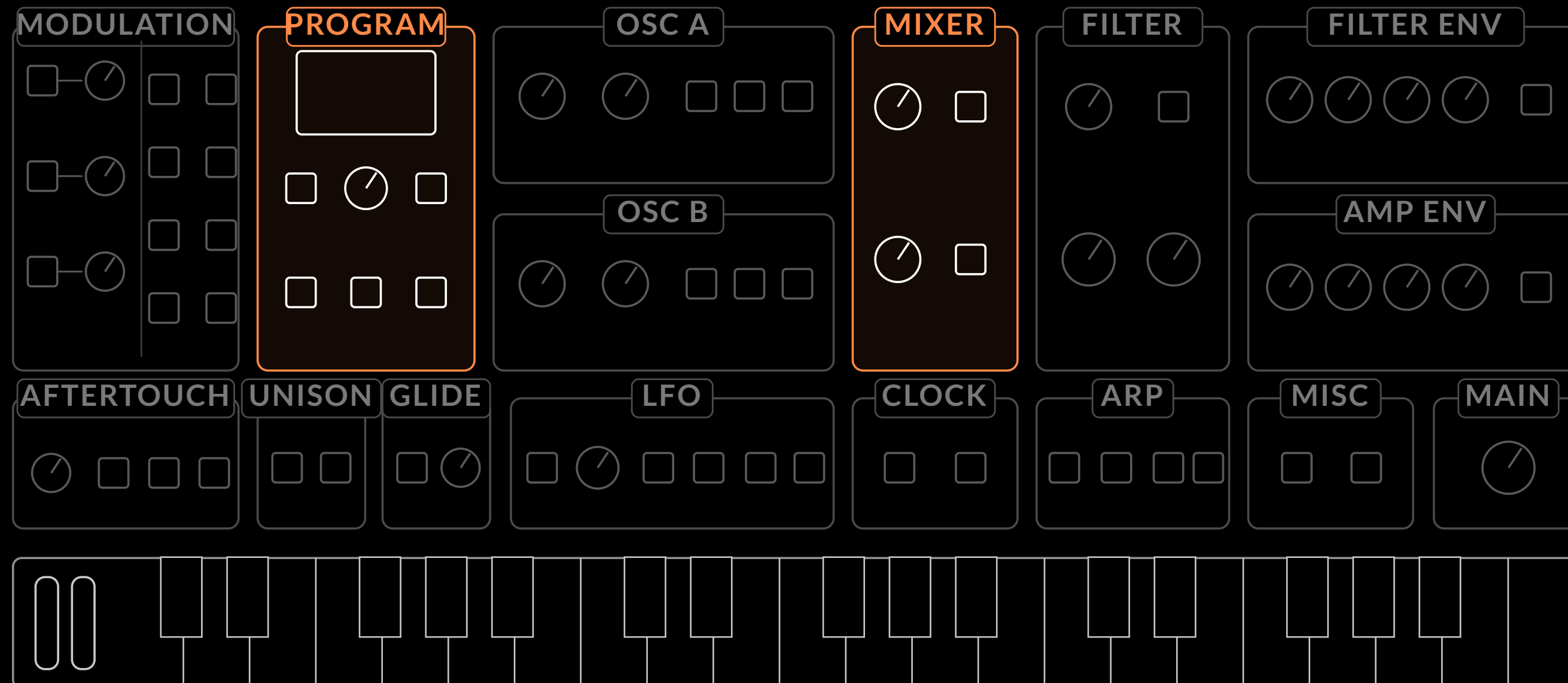


DEFAULT LEVEL 100 • HOLD FEEDBACK + TURN SELECT • THICKEN, DRIVE, CHAOS

WHERE ON FOURM

MIXER › FEEDBACK + SELECT

FEEDBACK on; hold it and turn SELECT for the amount: warmth first, then growl.



KEY TERMS

New words from this module

UNISON

All chosen voices play the same note. Monophonic, massive.

CHORD MEMORY

Unison stores a chord; single keys replay it transposed.

FEEDBACK

The output fed back into the mixer, soft-clipped.

POLYPHONY • VOICE

Four complete voices; the fifth note steals the oldest.

UNISON DETUNE

0–127 of pitch offset between stacked voices: a thickening that sounds like a chorus.

HOLD

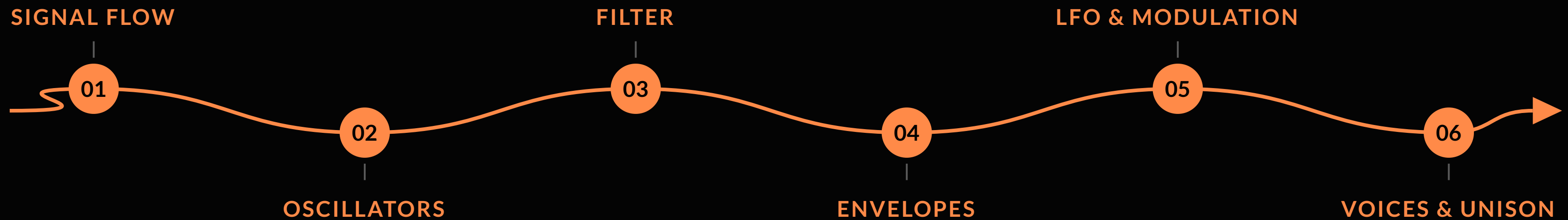
Sustains up to four notes after your hands leave, at the amp envelope's sustain level; also latches the arpeggiator.

SOFT-CLIPPING

Gentle rounding of peaks: saturation instead of harsh distortion.

THE ROAD, AGAIN

Every synth is this road with different clothes.



THANK YOU

Take the handbook home, EN or RO

SINTEZAUR

the community & what we build next



ZEEDO

try and buy the Fourm



SEQUENTIAL

user guide & presets



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GLOSSARY

The full glossary is in your handbook

